Suppression scenario laboratory: case study and failure modes

Status: public methods note. Not an operational fire-suppression model.

1. What this is

FRESH Fire Studio is a public 3D benchmark-fire portal from the UBC FRESH Lab. The suppression add-on asks a narrow software and methods question:

> Given published suppression evidence and a public fire benchmark, what can an open, auditable scenario model show, and where does it fail?

It is not a forecast of what a real suppression operation would achieve.

2. Evidence base

We use published evidence rather than a bespoke field campaign.

Wheatley (2024), Chapter 5 - IR-monitored experimental burns:

- mean ROS reduction: **0.5 ± 0.7 m/min**
- mean fireline-intensity reduction: **72 ± 159 kW/m**
- mean suppression hold: **123 ± 117 s**
- high-water hold: **187 s**
- indirect high-water hold: **243 s**
- direct water depths: **1.4-2.3 mm**
- indirect water depths: **0.7-1.1 mm**

McFayden et al. (2023) - aerial drop guide:

- effective aerial drop hold: **60 min**
- nested water-depth contours, stacking and breach criteria
- operational intensity discussion up to **10,000 kW/m**

Van Wagner & Taylor (2022) - water requirements:

- theoretical depth for quenching, cooling and wetting as a function of fireline intensity, fuel load and rate of spread.

The benchmark is CFSDS `2003_526`: 520 ha observed, 5,325 ha free-burn model, public fuel/terrain, Open-Meteo/FireSTARR reanalysis weather.

3. What we built

- a native time-dependent containment adapter;
- an evidence-bounded uncertainty sampler;
- a 32-member ensemble;
- a realistic 21-target, three-aircraft cycle mission;
- a synthetic control barrier;
- dispatch, queueing, sub-cell drop geometry and water-depth modules;
- a public diagnostic replay at `/suppression/`.

4. Results

Case	Area	Meaning
Observed final extent	**520 ha**	public CFSDS perimeter
Free burn	**5,325 ha**	model basin-filling result
Realistic cycle mission	**5,325 ha**	all 32 evidence samples; 0% containment
Synthetic control barrier	**3,574.7 ha**	3,570-3,575 ha; ROS correlation -0.78

The 21 sorties deliver 126,000 L. Covering the 5.3 km selected head segment at 0.75 mm would require about **199,630 L**; the line-mean depth is **0.47 mm**, coverage is **39%**, and the deficit is about **73,630 L**, or **13 additional sorties**. A two-pass case delivers **252,000 L**, reaches **0.95 mm** line-mean depth but still covers only **39%** of the target strip because it repeats the same 21 cells; the mass deficit closes, while the native outcome remains **5,325 ha**. A continuous 54-target line delivers **324,000 L** at 100% target coverage and still ends at **5,325 ha**. The realistic mission is therefore both placement-limited and water-limited.

The realistic mission is **placement-limited**. The control barrier is a deliberately permanent, 20-cell-wide synthetic wet band. It is not an operational plan. Its purpose is to test whether the evidence branch responds when the geometry is sufficient.

It does: the control reduces area by about 1,750 ha and the sampled absolute ROS reduction has a strong negative correlation with final area.

5. Failure modes

1. **Placement** - finite lines leave open flanks; the fire bypasses them.
2. **Resolution** - a 100 m cell cannot resolve a 50-100 m drop footprint.
3. **Transferability** - the experiments are low-intensity surface fires, not `2003_526`.
4. **Effect mapping** - the water-depth/ROS mapping is diagnostic.
5. **Free-burn bias** - the model burns 10x the observed area before suppression.
6. **Time** - water decays; a single mission is not a sustained response.

6. What would change the conclusion

- a realistic wet barrier that intersects the active front continuously;
- a validated water-depth-to-fuel-moisture/ROS relationship;
- a finer fire grid with source fuel data;
- an operational partner with a specific decision and data-sharing agreement.

A field campaign is not a prerequisite for publishing the software or the failure-mode analysis. It is a prerequisite for an operational claim.

7. Reproduce

- replay: https://ubc-fresh.github.io/fresh-fire-studio/suppression/
- model card: `docs/p179-18-model-card.md`
- evidence notes: `docs/martell-pdf-notes.md`
- ensemble: `suppression/proofs/2003-526-evidence-ensemble.json`
- control: `suppression/proofs/2003-526-evidence-control.json`

No operational use. No regional calibration. No claim of measured suppression effectiveness for `2003_526`.